MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. A normal vector to 2 $x - 3y + 1 = 0$ is

A. $(3; 2)$
B. $(2; -3)$
C. $(2; 3)$
D. $(1; 0)$

2. A direction vector of $x + 2y = 0$ is

A. $(-2; 1)$
B. $(1; 2)$
C. $(2; 1)$
D. $(0; 2)$

3. Centre of $x + 2)^2 + (y - 3)^2 = 4$:

A. $(2; 3)$
B. $(-2; -3)$
C. $(3; -2)$
D. $(-2; 3)$

4. Radius of $x^2 + y^2 = 9$:

A. 9
B. 81
C. 3
D. $\sqrt{3}$

5. $x^2 + y^2 = -1$ describes

A. a circle
B. the empty set
C. the origin
D. a line

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. Projection of (3; 4) onto the x-axis:

A. $(0; 4)$
B. $(0; 0)$
C. $(3; 0)$
D. $(4; 3)$

2. Distance from (1; 5) to $y = 2$:

A. 3
B. 5
C. 1

D. $\sqrt{10}$

3. To define $ax + by + c = 0$ to be a line, we need

A. $a \neq 0$ only

B. $b \neq 0$ only

C. $c \neq 0$

D. $(a; b) \neq (0; 0)$

4. $x^2 + y^2 = 0$ describes

A. a circle of radius 1

B. the single point O

C. a line

D. the plane

5. Circle $x^2 + y^2 = 4$ and the line $x = 2$: number of common points

A. 0

B. 2

C. 1

D. Infinite

True or false? Justify.

1. The orthogonal projection of P onto a line is unique.

2. Every equation $x^2 + y^2 + c = 0$ describes a circle of positive radius.

3. A line can meet a circle at three distinct points.

Personal review

I give a non-zero normal vector; I check the projection; I complete both squares; I distinguish circle, point and empty set.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.